

Euclid-O-Matic v0.2

a 4 Output Euclidean Trigger / Gate Sequencer



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Description

Euclid-O-Matic is a four channel trigger sequencer that is capable of creating euclidean rhythm patterns. Each pattern can be 1 to 16 steps. The euclidean patterns can be used to trigger e.g. percussion voices. When choosing different step numbers per pattern one can easily create phase shifting polyrhythms for e.g. Berlin style melodic sequences. 16 four channel patterns can be saved and recalled. The triggers / gates can be one or more clock pulses long. When 1 clock pulse long you can choose the pulse width of the output gate via the pulse width control. The gates are combined in an OR output and send to a 16 step DAC which can be used as a synchronous LFO. This manual describes Euclid-O-Matic software version 1.6 running on hardware version 0.2. You can find code, schematics, 3D design files and gerbers at <https://github.com/josbouten>

Getting started

Connect a clock signal to the Clock In input. Only after connecting a clock you will see trigger signals on the outputs and patterns on the pixel ring and LEDs light up. The clock pulse can be divided by a factor 1, 2, 4, 8, 16, 24, 32, 48, 64 or 96. Choose the appropriate factor for your setup pressing the Clock Division Button. This factor will be stored in EEPROM, so next time you fire up Euclid-O-Matic it starts with the factor you chose. See also the paragraph “External Clock” below.

LED test and software version number

Upon startup the sequencer will show all LEDs in an alternating black and white pattern so that you can see that they are all working. It will then flash the major.minor version number of the software in green and in red a few times. The mode LED will show the 3 mode colours. Finally the last channel used is chosen. Once a clock is connected the corresponding pattern is shown. A running cursor will point to the current position in the pattern. Without a clock signal you will not see a pattern on startup.

External Clock

The sequencer will run on an internal clock. Connect an external clock to the ‘clock in’ input. Press the ‘Divide’ button to reduce the clock rate by stepping through a list of reduction factors 1, 2, 4, 8, 16, 24, 32, 48, 64 or 96. By pressing the button multiple times you select a higher division factor. Double pressing the push button will step back through the list. The value chosen will be stored in the EEPROM whenever the rotary dial is rotated or a trigger button is pressed. The LED in the push buttons will blink whenever the division factor is not the same as when the module was started otherwise it will be lit continuously.

Cursor

A cursor will show the current step position in the pattern. It will change colour depending on the trigger channel selected.

Trigger Channel

You can choose either trigger channel A, B, C or D by pressing the corresponding push button. When pushed you will see information about the pattern chosen for that trigger channel, you can change the pattern length, the trigger pattern and the positions it will trigger in relative to the patterns of the other trigger channels. The LEDs of the circular display will light up in the colour of the trigger channel (red, blue, green or yellow).

Mute a channel

Pressing the 'Shift Button' and either trigger button A, B, C or D will toggle a mute state for that trigger channel.

Modes

Euclid-O-Matic has 4 modes. A 'Trigger Count Mode', a 'Pattern Length Mode', a 'Rotate Mode' and a 'Program Mode'. You can toggle between 'Trigger Count Mode' and 'Pattern Length Mode' by double clicking the rotary encoder push button. You can toggle between 'Trigger Count Mode' and 'Rotate Mode' by single clicking the rotary encoder push button. You can toggle between 'Pattern Length Mode' and 'Rotate Mode' by single clicking the rotary encoder push button.

Using the 'Trigger Count Mode' you can choose the number of triggers in a pattern. In all modes except the Program Mode you can press and hold the 'Shift Button' and increase or decrease the duration of the output trigger signal by clock pulse steps using the rotary encoder turning it into a gate signal. You might e.g. change a 4 trigger euclidean pattern where each trigger is one clock pulse long to 4 gates where each of them lasts 3 clock pulses as long. If you use this type of gates to trigger a synth voice and use multiple channels with different number of steps per pattern you can make interesting polyrhythms (as used e.g. in early Berliner Schule style of music).

You can switch to 'Program Mode' by long pressing the rotary encoder push button.

Trigger Count Mode

At first startup the sequencer will be in 'Trigger Count Mode', so the status led will be green. For any of the 4 trigger patterns you can set the number of trigger pulses for its pattern. Rotating the encoder will increase or decrease the number of pulses in the pattern for the chosen trigger channel. Each time you change the number of pulses a suitable Euclidean pattern is generated. The LEDs above the 4 outputs show the ON/OFF status of the respective gate.

Gate Length

To change the gate length, press the 'Shift Button' and turn the rotary encoder. You can do this in any mode except the 'Program Mode'. The resulting pattern will be a euclidean one. You can make a single gate as long as 15 steps. If the gate length is one then you can set its pulse width. You can change the pulse width using the 'Pulse Width' dial from 5% to 95%. Note that if you have a trigger count and increase the 'Gate Length', then the effective trigger count is decrease if there are not enough steps available to increase the length for each gate. If you decrease the gate length enough then the original trigger count will be restored.

Pattern Length Mode

When switched to the 'Pattern Length Mode' the mode led will be blue. In this mode you can set the total number of steps in the pattern. Initially this will be 16 but you can choose any value from 1 up to 16. When the length of the pattern is smaller than 16 steps you will see some unlit positions on the led-ring. If you want the channel to never trigger, use the mute function.

Rotate Mode

In either 'Trigger Count Mode' or 'Pattern Length Mode' you can press the rotary button once and switch to the 'Rotate Mode' (pressing once more will bring the device back to the previous mode). In 'Rotate Mode' the mode led will light up orange. it is possible to shift the trigger pulse pattern across the total length of the pattern (with wrapping) for the trigger output selected. In this way you can set the moment a channel will send a trigger in relation to the other channels.

Program Mode

Long pressing the rotary encoder's push button will select the 'Program Mode'.

There are 16 memory locations available. The LEDs will light up in RED for the memory locations already containing a pattern and in GREEN for the locations that are empty. Using the rotary encoder's knob one of 16 memory cells can be selected. For each trigger channel all information is stored, including its mute state.

Pressing trigger channel A's button will recall the patch pattern stored in the EEPROM memory cell and store it in the sequencer's active memory from the patch number selected with the rotary encoder.

Pressing trigger channel B's button has no effect.

Pressing trigger channel C's button will clear the patch location the rotary encoder is set to. The pattern already in the working memory however will not be changed. Don't forget to save it if you like it.

Pressing trigger channel D's button will store the current pattern of the selected trigger channel to the corresponding memory cell.

You can leave this mode and return back to the prior mode by pressing the rotary encoder's push button once.

The pattern information is stored in an EEPROM and retained when the device is power cycled. For each pattern the mute status of the channels is stored as well.

Wipe all patterns from memory

Should you want to wipe all patterns from memory and return to the factory preset, then press the buttons in the following succession: A D B C A D B C. The pixel circle will then show a blinking chequered pattern in red and white and all patches will be wiped in an EEPROM friendly way. There is no undo!

Mode Colours

The mode the Euclid-O-Matic is in is always visible via the Mode Led. It changes colour depending on which mode is active.

Green	Trigger Count Mode
Blue	Pattern Length Mode
Red	Rotate Mode
White	Program Mode
Flashing	Currently Saving Settings to EEPROM

Whenever the mode LED is flashing the device is storing data to the EEPROM inside its processing unit. Make sure not to switch off the device when the LED is flashing. If you do, the present settings are not saved properly and the device will revert to the settings that were saved successfully the last time. To prevent saving the data to EEPROM to interfere with the timing of the trigger pulses it is done in small chunks at a low priority by the MPU. It will take about 10 seconds to finish.

Inputs / Outputs

F-1, F-2

These inputs can be used to shift the pattern of channel A and B. A low to high transition on the input will make the pattern shift counter clockwise one step. If the output of channel A or B is connected to F-1 or F-2, each pattern step will trigger the corresponding output and a rotation step which may lead to a very fast series of gate signals.

STOP/START input

This input can be used to stop or start the sequencer with a logic signal. A continuous low signal will stop the sequencer, a high signal will start it. Note, when no signal is connected to the sequencer it will start running at startup.

4OR output

This output fires whenever one of the trigger channels produces a high on its output. In other words, it is a logic OR of all trigger channel outputs.

LFO output

The signals sent to the four gate outputs are also used as the input bits of a DAC. This signal is available at the LFO-output. Given that it is a combination of all 4 gate outputs, it will produce a cyclic pattern that may be used as an LFO of sorts. Trig A is used as the most significant bit, trig D is the least significant bit.

Configuration

Several choices were made to make the software work as it does. Some are determined by the constants defined in the file “config.h”. If you want to make changes to these choices have a look at that file. Whenever you change a value don’t forget to recompile the code and upload it to the arduino nano Euclid-O-Matic uses as its processing unit. For example the Trig A output pattern will respond to a pulse on the F-1 input. You can change the rotation direction via the ROTATE_CLOCKWISE_ON_FN1 constant.

EEPROM_WRITE_DELAY_TIME

Whenever you adjust a setting it will be stored in EEPROM so that the device will retain that setting on startup or you can select a patch in program mode from the 16 memory locations. Writing to the EEPROM is delayed so that if you make several changes within a short while they will not be written to the EEPROM immediately as single changes but as a combined group of changes. The delay time can be set using the EEPROM_WRITE_DELAY_TIME value in the “config.h” file. Save the file, compile the source code and upload the binary.

Design and Code repository

You can find code, schematics, 3D design files and gerbers at <https://github.com/josbouten>

Frequently Asked Questions (faq)

Q: Why is there no pattern visible after the module is connected to a power supply?

A: You must supply a clock signal to the clock input. As soon as a rising pulse is detected a pattern will be shown and the LEDs above the trigger outputs will start to blink.

Q: Where can I read more about Euclidean Rhythms?

A: You may want to start with a paper by Godfried Toussaint titled: “The Euclidean Algorithm Generates Traditional Musical Rhythms”.

You can find it here: <https://cgm.cs.mcgill.ca/~godfried/publications/banff.pdf>